MD SHAHRIAR SHAKIR SUMIT

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LinkedIn ◇ Google Scholar ◇ ORCID

SUMMARY

I am a final-year PhD candidate in the School of Engineering and Technology at UNSW Canberra, with thesis submission expected in December 2026. My research focuses on computer vision, object detection and trustworthy AI, addressing challenges in occlusion, low-light environments and resource-constrained deployment. My broader research interests include multimodal foundation models, test-time adaptation, federated learning, reinforcement learning, world models and medical imaging. I am committed to advancing impactful research, interdisciplinary collaboration and excellence in university teaching. "'

EXPERIENCE

University of New South Wales (UNSW Canberra), Canberra, Australia *May 2023 – Present*
PhD Researcher, School of Engineering and Technology
Research on robust and efficient object detection in complex scenes, including occlusion, illumination variation and resource-constrained deployment.

American International University-Bangladesh, Dhaka, Bangladesh *Apr. 2022 – Mar. 2023*
Visiting Research Assistant
Developed convolutional neural networks for accurate real-time human detection.

Universiti Teknologi PETRONAS, Perak, Malaysia *Oct. 2018 – 2021*
Graduate Assistant and Research Assistant
Conducted research on human detection using convolutional neural networks and supported research activities.

EDUCATION

University of New South Wales (UNSW Canberra), Canberra, Australia *2023 – Present*
Doctor of Philosophy in Computer Vision, School of Engineering and Technology
Research: Vision-Based Deep Neural Networks to Improve Object Detection Performance in Complex Scenes
PhD Supervisor: Dr Murat Tahtali

Universiti Teknologi PETRONAS, Perak, Malaysia *2018 – 2021*
Master of Science by Research in Computer and Information Sciences (Computer Vision)

American International University-Bangladesh, Dhaka, Bangladesh *2011 – 2016*
Bachelor of Science in Computer Science and Software Engineering

SELECTED AWARDS AND HONOURS

- **Top Reviewer**, Advances in Neural Information Processing Systems (NeurIPS) 2026
- **Outstanding Reviewer**, 36th British Machine Vision Conference (BMVC) 2025
- **Fourth place**, Three Minute Thesis (3MT) Competition, UNSW Canberra 2025
- **Best Presentation Award**, SET & SYSCOM Conference, UNSW Canberra 2024
- **Tuition Fee Scholarship and Research Stipend**, UNSW Canberra 2023–2026
- **Meritorious Master by Research Student Award**, Universiti Teknologi PETRONAS 2021
- Graduate Research Assistant, Yayasan UTP Grant 2020–2021
- Graduate Assistant, Centre for Graduate Studies, Universiti Teknologi PETRONAS 2018–2020

RESEARCH PAPER PUBLICATIONS

Major Research Topics:

- Computer Vision, Pattern Recognition, Object Detection and Image Classification
- Efficient and Robust AI for Occlusion, Low-Light Conditions and Edge Deployment
- Multimodal Foundation Models, Vision-Language Models and Test-Time Adaptation
- Federated Learning, Reinforcement Learning, World Models and Trustworthy AI Evaluation

RESEARCH PAPER PUBLICATIONS

Major Research Topics:

- Computer Vision, Pattern Recognition, Object Detection and Image Classification
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- Multimodal Foundation Models, Vision-Language Models and Test-Time Adaptation

- Federated Learning, Reinforcement Learning, World Models and Trustworthy AI Evaluation

Recent Publications

[P7] [WACV'27] S. S. Sumit, M. Tahtali, and S. Mirjalili, "AAA-Bench: Counterfactual, Verifier-Grounded Evaluation of Vision-Language Models as Decision-Makers Under Visual Ambiguity," IEEE/CVF Winter Conference on Applications of Computer Vision, 2027. (accepted)

[P6] [WACV'27] S. S. Sumit, M. Tahtali, and S. Mirjalili, "Visible-Region Attention Supervision for Occluded Pedestrian Detection," IEEE/CVF Winter Conference on Applications of Computer Vision, 2027. (accepted)

[P5] [ACML'26] S. S. Sumit, M. Tahtali, and S. Mirjalili, "MEI-Bench: Diagnosing Causal Visual Grounding in Vision-Language Models via Minimal Evidence Interventions," 18th Asian Conference on Machine Learning, 2026. (accepted)

[P4] [ACML'26] S. S. Sumit, [coauthors], "When TTA Meets Frozen Detectors: A Benchmark and Feature-Anchoring Solution for Streaming Object Detection," 18th Asian Conference on Machine Learning, 2026. (accepted)

[P3] [KES'26] S. S. Sumit, M. Tahtali, D. Byrne, S. Mirjalili, A. Miroshnichenko, and M. Garratt, "HOD-Lite: An Enhanced Lightweight Human Object Detection Network via Neural Architecture Search, Knowledge Distillation and Pruning," 30th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems, 2026. (accepted)

[P2] [KES'24] S. S. Sumit, S. Anavatti, M. Tahtali, S. Mirjalili, and U. Turhan, "ResNet-Lite: On Improving Image Classification with a Lightweight Network," *Procedia Computer Science*, 246, 1488–1497, 2024.

[P1] [ACIIDS'24] S. S. Sumit, S. Anavatti, M. Tahtali, S. Mirjalili, and U. Turhan, "KDVGG-Lite: A Distilled Approach for Enhancing the Accuracy of Image Classification," LNCS 14796, 15–27, Springer, 2024.

Prior to PhD

[P5] [MethodsX'23] S. S. Sumit, D. R. A. Rambli, S. Mirjalili, M. S. U. Miah, and M. M. Ejaz, "ReSTiNet: An Efficient Deep Learning Approach to Improve Human Detection Accuracy," *MethodsX*, 10, 101936, 2023.

[P4] [Applied Sciences'22] S. S. Sumit, D. R. A. Rambli, S. Mirjalili, M. M. Ejaz, and M. S. U. Miah, "ReSTiNet: On Improving the Performance of Tiny-YOLO-Based CNN Architecture for Applications in Human Detection," 12(18), 9331, 2022.

[P3] [Springer'21] S. S. Sumit, J. Watada, F. Nasrin, N. I. Ahmed, and D. R. A. Rambli, "Imputing Missing Values: Reinforcement Bayesian Regression and Random Forest," in *Soft Computing for Biomedical Applications and Related Topics*, 81–88, 2021.

[P2] [IEEE Access'21] S. S. Sumit, D. R. A. Rambli, and S. Mirjalili, "Vision-Based Human Detection Techniques: A Descriptive Review," 9, 42724–42761, 2021.

[P1] [JPCS'20] S. S. Sumit, J. Watada, A. Roy, and D. R. A. Rambli, "In Object Detection Deep Learning Methods, YOLO Shows Supremum to Mask R-CNN," *Journal of Physics: Conference Series*, 1529(4), 042086, 2020.

Manuscripts Under Review

[U1] S. S. Sumit, M. Tahtali, and S. Mirjalili, "HOD: A Review of Human Object Detection Through the Ages: From Handcrafted Features to Deep Learning." Under review at *Engineering Applications of Artificial Intelligence*.

[U2] S. S. Sumit, M. Tahtali, and S. Mirjalili, "When the Metric Cannot See the Failure: Silent Evaluation Faults in Byzantine-Robust Federated Learning." Under review at *Transactions on Machine Learning Research*.

[U3] S. S. Sumit, M. Tahtali, and S. Mirjalili, "Human Object Detection from Handcrafted Features to Foundation Models: Systematic Literature Review." Under review at *ACM Computing Surveys*.

[U4] S. S. Sumit, M. Tahtali, and S. Mirjalili, "A World Model Can Predict the Future Without Knowing Its Cause: Interventional Evaluation of Action-Conditioned World Models." Under review at *International Conference on Learning Representations*.

[U5] S. S. Sumit, M. Tahtali, S. Mirjalili, and A. Miroshnichenko, "Robust Aggregation Prefers Data-Poor Clients: A Sample-Size Bias in Distance-Based Byzantine Defences." Under review at *International Conference on Artificial Intelligence and Statistics*.

[U6] S. S. Sumit, M. Tahtali, S. Mirjalili, and A. Miroshnichenko, "Marginal Coverage Is an Unsafe Diagnostic for Sparse Events." Under review at *International Conference on Artificial Intelligence and Statistics*.

[U7] S. S. Sumit, M. Tahtali, S. Mirjalili, and A. Miroshnichenko, "When Does Geometric Screening Fail in Federated Reinforcement Learning? The Dispersion Trap, and a Screened-Mean Aggregator That Respects It." Under review at *International Conference on Artificial Intelligence and Statistics*.

Manuscripts in Preparation

[W1] S. S. Sumit, M. Tahtali, and S. Mirjalili, "ILL-GUARD: Byzantine-Resilient Federated Learning for Nighttime Pedestrian Detection via Illumination-Aware Layered Defense." In preparation for CVPR.

[W2] S. S. Sumit, M. Tahtali, and S. Mirjalili, "Depth and Medium Share One Gauge: Profiling a Distributed Medium at Unknown Range." In preparation for CVPR.

[W3] S. S. Sumit, M. Tahtali, and S. Mirjalili, "Auditing Federated Unlearning in Medical Imaging: What Membership Inference Cannot Tell You, and What Can." Manuscript in preparation.

TALKS/POSTER PRESENTATIONS

- **[T3] [KES'26]** "HOD-Lite: An Enhanced Lightweight Human Object Detection Network," accepted oral presentation at KES — 2026
- **[T2] [3MT'25]** "Lightweight AI: Detecting People When Every Second Counts," Three Minute Thesis Competition, UNSW Canberra — 2025
- **[T1] [SET & SYSCOM'24]** Research presentation, UNSW Canberra; Best Presentation Award — 2024

TEACHING EXPERIENCE

- **Lecturer (Casual)**, Crown Institute of Higher Education, Canberra — Aug. 2023 – Present
- **Tutor, IT Professional & Ethics (TEB2173)**, Universiti Teknologi PETRONAS — May 2020 – Sep. 2021
- **Tutor, Computer Ethics & Cyber Law (TEB3233)**, Universiti Teknologi PETRONAS — May 2021
- **Tutor, Fundamentals of Information Systems (TEB1153)**, Universiti Teknologi PETRONAS — Jan. – Sep. 2019

SERVICE

- Peer reviewer, *Artificial Intelligence Review*, *Information Sciences*, *Cluster Computing*, *Scientific Reports*, *IEEE Access*, *Signal, Image and Video Processing*, and *Discover Artificial Intelligence* — Ongoing
- Outstanding Reviewer, British Machine Vision Conference (BMVC) — 2025

GRANTS AND RESEARCH FUNDING

- Development and Research Training Grant, UNSW: AUD 3,000 — 2026
- UNSW HDR Funding for conference attendance and research activities: AUD 5,000 — 2023
- NCI Australia high-performance computing allocations: 81 KSUs (Q4 2024), 43 KSUs (Q1 2025), 81 KSUs (Q2 2025), 90 KSUs (Q3 2025), 110 KSUs (Q4 2025), 143 KSUs (Q1 2026), 200 KSUs (Q2 2026) and 1.41 MSUs (Q3 2026) — 2024–2026

PROFESSIONAL MEMBERSHIP AND REGISTRATION

- Graduate Technologist, Malaysia Board of Technologists (MBOT) — 2024 – Present

TECHNICAL SKILLS

Programming and ML: Python, PyTorch, TensorFlow, Keras, Java, C, C++ and .NET.
Web and Data: HTML5, CSS3, JavaScript, jQuery, Bootstrap, PHP, SQL and MySQL.
Research Tools: LaTeX, Git, Microsoft Word and high-performance computing environments.

RESEARCHER IDENTIFIERS

Google Scholar: 169 citations; h-index 5; i10-index 3 | **ORCID:** 0000-0002-7435-9214
Scopus ID: 57218118363 | **ResearcherID:** U-2534-2017

REFEREES

Dr Murat Tahtali — *murat@unsw.edu.au*
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